

CHAPTER 21

PROGRAM DESIGN AND TECHNIQUE FOR AEROBIC ENDURANCE AND METABOLIC TRAINING

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The authors would like to acknowledge the significant contributions of Benjamin H. Reuter, J. Jay Dawes, and Patrick S. Hagerman to this chapter.

After completing this chapter, you will be able to

- discuss the factors related to aerobic endurance performance;
- select the mode of aerobic endurance training;
- set aerobic endurance training frequency based on training status, sport season, and recovery requirements;
- assign aerobic endurance training duration and understand its interaction with training intensity;
- assign aerobic endurance exercise intensity and understand the various methods used to monitor intensity;
- describe the various types of aerobic endurance programs;

- apply the program design variables based on the sport season; and
- address the issues of cross-training, detraining, tapering, concurrent training, heat acclimation, and altitude when designing an aerobic endurance training program.

This chapter discusses the general principles of program design as they apply to aerobic endurance training and a stepwise approach to designing a safe and effective program.

Improvements in aerobic endurance performance can be derived only when sound principles are applied during training. While some of the fundamental mechanisms responsible for inducing adaptations during training are undefined, it is clear that in order to adapt, the various systems and tissues of the body must be challenged by an exercise stimulus (e.g., specificity and overload). The physiological systems that are not involved during the training session or not stressed sufficiently by exercise will not adapt to the training program (5, 85).

Training specificity refers to the distinct adaptations to the physiological systems that arise from the training program. Moreover, a training effect is limited to the physiological systems used and overloaded during training (140, 142). To improve aerobic endurance performance, training programs must be designed to enhance the function of the respiratory, cardiovascular, musculoskeletal, and nervous systems.

For a training adaptation to occur, a physiological system must be stressed at a level beyond which it is presently accustomed (105). Following repeated and consistent overload, the physiological systems of the body adapt to the exercise stress. Adaptations occur until the tissues are no longer overloaded by the applied exercise stress. This necessitates the use of a greater overload for continued adaptations. Exercise frequency, duration, and intensity are the variables most often manipulated to provide overload to the systems and tissues of the body.

Successful performance in aerobic endurance competitions involving running, cycling, and swimming depends on athletes' ability to cover a fixed distance in the shortest time possible. This requires athletes to be in peak physical and psychological condition for the competition. To reach this level of performance, athletes must consistently train hard, yet intelligently, to maximize the physiological adaptations derived from training while avoiding injury and overtraining. A commonly observed trend with aerobic endurance athletes is to adopt and embrace the training practices of highly successful or well-known athletes. Although this strategy may be effective for a few, most aerobic endurance athletes would likely be better served by constructing their own training regimen based on a good working knowledge of sound training principles and an understanding of their own individualized physical limitations and needs. Another common belief is that greater overload via longer duration, higher intensity, and greater frequency is required to improve performance. However, training smarter by identifying physiological, biomechanical, and performance strengths and weaknesses (e.g., poor leg strength, muscle imbalances, low heat tolerance) and working to consolidate strengths while correcting weaknesses, has greater opportunities to improve performance outcomes with lower risk of injuries. An extension of this principle is applying the **minimum effective dose (MED)** (132, 133). MED was originally used in the pharmaceutical industry to reflect the understanding that there is a minimum dose of medication needed to elicit the desired physiological response; the use of medications results in varying side effects, and the higher the dose of medication, the greater the side effects. In strength and conditioning, this term has been adapted to applying the minimum amount of training stimuli required to maximize adaptations while simultaneously minimizing the damage and injury risks associated with training. This concept has also been explored as the minimum training dose needed to maintain the adaptations of resistance training (126). In conjunction with MED, aerobic endurance coaches

and athletes have also come to understand the **maximum tolerated dose (MTD)** or maximum effective dose (133). This can be considered the maximum training workload that provides measurable benefits to the athlete. However, as discussed earlier, while still providing benefits, the side effects of MTD training stress (e.g., tissue microtrauma and fatigue) are also relatively high, and the risk–reward balance must be strongly considered.

This chapter is designed to provide the strength and conditioning professional with a good working knowledge of the scientific principles of aerobic endurance training and conditioning. Specifically, this chapter includes information about the factors related to performance, aerobic endurance training program design variables, and various types of training programs. Additional discussion focuses on sport season training and special issues related to aerobic endurance training including concurrent training (i.e., combining resistance and aerobic endurance training in one training program) and heat acclimation. Because it would be exhaustive to summarize all the relevant information about training for all possible aerobic endurance sports, only foundational aerobic endurance training topics are presented, with specific examples as they relate to running, cycling, and swimming.

Factors Related to Aerobic Endurance Performance

When designing aerobic endurance training programs, it is important to understand those factors that influence and play a significant role in successful aerobic endurance performance. This allows for the development of sound training programs while minimizing unnecessary training that may lead to counterproductive adaptations, fatigue, injury, or overtraining.

Maximal Aerobic Capacity

As the duration of the aerobic endurance event increases, so does the proportion of the total energy demand that must be met by aerobic metabolism. Therefore, high **maximal aerobic capacity ($\dot{V}O_{2\max}$)** is necessary for success in aerobic endurance events (87). A high correlation has been shown to exist between $\dot{V}O_{2\max}$ and performance in aerobic endurance events (2, 3, 134, 87). Consequently, aerobic endurance training programs should be designed, in part, to improve $\dot{V}O_{2\max}$. However, although a high $\dot{V}O_{2\max}$ is important for successful performance, other factors may be equally or even more important. These factors include a high lactate threshold, good exercise economy, high efficiency in using fat as a fuel source, and a high percentage of type I muscle fibers.

For well-trained aerobic endurance athletes, improving $\dot{V}O_{2\max}$ may benefit performance only up to a certain point, especially since these individuals typically already possess excellent aerobic capacity. Consequently, the ability to sustain higher velocities during competition and training may have a greater impact on performance than attempting to make marginal improvements in aerobic capacity. For this reason, many athletes use high-intensity interval training (HIIT). HIIT may contribute to performance in highly trained aerobic endurance athletes via improvements in peak power output,

ventilatory threshold, hydrogen ion buffering, use of fat as a fuel source, and improved exercise economy (76).

Lactate Threshold

In aerobic endurance events, the best competitor among athletes with similar $\dot{V}O_2\text{max}$ values is typically the person who can sustain aerobic energy production at the highest percentage of his or her $\dot{V}O_2\text{max}$ without accumulating large amounts of lactic acid in the muscle and blood (74). Although numerous terms have been used to refer to this phenomenon, **lactate threshold (LT)** is the one most commonly employed in the literature. Lactate threshold is that speed of movement or percentage of $\dot{V}O_2\text{max}$ at which a specific blood lactate concentration is observed (i.e., 4 mmol/L of blood) or the point at which blood lactate concentration begins to increase above resting levels (136). Several studies have shown that an athlete's LT appears to be a better indicator of his or her aerobic endurance performance than $\dot{V}O_2\text{max}$ (7, 113). The **maximal lactate steady state (MLSS)** is another term that often appears in the aerobic endurance training literature. The MLSS is defined as the exercise intensity at which maximal lactate production is equal to maximal lactate clearance within the body (14). MLSS is an indicator of the transition from heavy to severe exercise intensity (70). In other words, while lactate production increases at the onset of exercise in the heavy exercise intensity domain, it plateaus and can be maintained at a steady state at this elevated level. However, if exercise intensity increases further, now entering the severe domain, lactate increases continually until failure. Notably, MLSS is also synonymous with *critical power* (CP) and *critical speed* (CS) in this regard. The MLSS is considered by many to be a better indicator of aerobic endurance performance than either $\dot{V}O_2\text{max}$ or the LT (14). What is clear from this information is that aerobic endurance athletes must improve their LT and MLSS. This requires athletes to conduct some training at elevated levels of blood and muscle lactate to maximize training improvements in these areas.

Exercise Economy

A measure of the energy cost of activity at a given exercise velocity is referred to as **exercise economy**. Athletes with a high exercise economy expend less energy during exercise to maintain a given exercise velocity (e.g., running speed). Several investigators have suggested that exercise economy is an important factor in successful performance in running events (71), with better performers also having a slightly shorter stride length and greater stride frequency compared to less successful performers (106). During cycling, exercise economy can be affected by body mass size, cycling velocity, and aerodynamic positioning (7, 41, 42). For cyclists, an increase in body mass and cycling velocity as well as an inefficient body position generate greater wind resistance, resulting in a decrease in exercise economy. It has been demonstrated that elite swimmers are much more economical than non-elite swimmers (6) and use less oxygen at any given swimming velocity. The most profound impact on exercise economy during swimming can be observed when swimming technique becomes more efficient. As improvements in stroke mechanics occur, energy demand for a given swimming velocity is reduced (146). Training to improve exercise economy is critical for aerobic endurance athletes.

An improvement in exercise economy can enhance maximal aerobic capacity ($\dot{V}O_{2\text{max}}$) and lactate threshold.

Designing an Aerobic Endurance Program

An effective aerobic endurance training program must include an exercise prescription specifically developed for the individual athlete based on his or her individual strengths and weaknesses, goals, and sport. This requires manipulation of the primary program design variables. The sidebar lists the design variables as steps 1 through 5. Unfortunately, strength and conditional professionals and athletes often use the training practices or programs of current successful strength and conditional professionals or athletes in their sport. This does not carefully consider the strengths and weaknesses of the athlete and may lead to development of an ineffective or potentially harmful training program. The optimal development of a safe and effective training program is to have the factors related to aerobic endurance performance evaluated and then use that information to generate a training program specific to the athlete. For example, an athlete with poor exercise economy should place a greater emphasis on training to improve exercise economy compared to other factors that are known strengths. This might include HIIT with a focus on technique, as well as using long rest periods. Conversely, athletes who need to increase LT might consider performing more high-intensity training with shorter rest periods.

Aerobic Training Program Design Variables

Step 1: Exercise mode

Step 2: Training frequency

Step 3: Training intensity

Step 4: Exercise duration

Step 5: Exercise progression

Training programs for female athletes do not have to be different from those used to train male athletes; evidence indicates that males and females respond similarly to training programs (12, 78). Refer to [chapter 8](#) for a discussion on sex-related differences and their implications for exercise.

Step 1: Exercise Mode

Exercise **mode** refers to the specific activity performed by the athlete: cycling, running, swimming, and so on. When training to improve aerobic endurance performance, the athlete should select activities that mimic as closely as possible the movement pattern employed in competition. This will cause positive adaptations in specific physiological systems and tissues of the body related to those used during competition. For example, the recruitment of specific muscle fibers and the adaptation of the energy systems within those fibers must be challenged during aerobic endurance training. Selecting the appropriate exercise mode during training ensures that the systems used in competition are challenged to improve. Note that the more specific the training mode is to the sport, the greater the improvement in performance is likely to be experienced. For an athlete involved in multiple aerobic endurance sports, or one who is interested in a general aerobic endurance fitness program, cross-training or participation in multiple aerobic endurance activities may be warranted (47).

Step 2: Training Frequency

Training **frequency** refers to the number of training sessions conducted per day or per week. The frequency of training sessions depends on an interaction of exercise intensity and duration, the training status of the athlete, and the specific phase of the sport (e.g., preparation phase or preseason versus competition phase or in-season). Higher exercise intensity and longer duration may necessitate less frequent training to allow sufficient **recovery** from exercise sessions. The training status of the athlete can influence

training frequency, with lesser-trained athletes requiring more recovery days at the beginning of a training period than more highly trained athletes. The sport phase or season that the athlete is currently in can also influence training frequency; an off-season program may include three to four training days per week, but training frequency may progress to daily workouts (or even multiple workouts per day for triathletes and highly competitive distance runners) in the preseason. Additionally, fewer training sessions may be required to maintain an achieved level of physiological function or performance than to attain that level initially (128). Appropriate training frequency is important for the aerobic endurance athlete, because too much training may increase the risk of injury, illness, or overtraining (i.e., maximum tolerable dose). While some evidence shows a positive relationship between exercise frequency and increased injury rates (63), these studies used active individuals in a wide age range, not only young and healthy athletes, as subjects. Additional research has shown injuries are multifactorial and there is insufficient evidence definitively associating injuries with increases or high levels of training frequency, volume, or other training factors (44). Conversely, too little training will not result in positive adaptations to the various systems of the body (i.e., below the minimum effective dose). Research has shown that it is necessary to train more than twice per week in order to increase $\dot{V}O_2\text{max}$ (138). Many strength and conditioning professionals recognize that multiple training sessions per day may be needed to improve performance in some aerobic endurance athletes. For athletes participating in single-discipline sports such as running, such training practices should be reserved for those with an advanced or elite training status due to the level of physiological stress involved with such programs. Research conducted by Hansen and colleagues (58) showed that time to exhaustion, resting muscle glycogen concentrations, and citrate synthase activity increased among seven healthy untrained men after 10 weeks of training twice every other day versus once daily. It was speculated that training in a glycogen-depleted state

may further increase glycogen resynthesis via an increase in the transcription and transcription rate of certain genes responsible for training adaptations. However, the researchers cautioned that these results should not necessarily be used by strength and conditioning professionals and practitioners to guide practice, because low glycogen concentrations may reduce the period of time an athlete is able to train and may increase risk of overtraining. This is one reason why it is important to monitor the effects of the training load on athletes. Further, strength and conditioning professionals and athletes need to understand that adaptation takes place during periods of rest, and undertaking more frequent training can affect the recovery process and, ultimately, the level of adaptation.

Recovery from individual training sessions is essential if the athlete is to derive maximum benefits from the subsequent training session. Exercise performance has been shown to improve following relative rest from difficult training sessions (16). Obtaining sufficient rest, becoming rehydrated, and restoring fuel sources are critical issues for the athlete during recovery. Physical and mental relaxation and avoidance of strenuous physical activity are particularly important following days of high-intensity or long-duration training. Postexercise ingestion of adequate fluids and electrolytes is important for replacing those lost during training. If the training session was especially long or intense, postexercise carbohydrate intake is important for replacing the muscle and liver glycogen stores that were likely depleted, as is the ingestion of protein to aid muscle repair and recovery. More detailed information on this topic can be found in [chapter 11](#).

Evidence suggests that while most aerobic endurance athletes across a variety of sports (e.g., cycling, running, walking, triathlon) use nutrition, hydration, stretching, and active recovery, other forms of recovery (e.g., compression garments and ice baths) are not used as frequently or at all (13, 19). In addition, while many athletes believe in the benefits of different recovery strategies for competition, such as active recovery, self-massage, and massage,

they do not actually use them (19). Research has further elucidated that elite athletes are more likely to use all forms of recovery, except cold water immersion, than lower tiered well-trained athletes (13). The use of or lack of use of different strategies has been found to be related to coach and athlete awareness of the effectiveness of such strategies (13). Further, the majority of athlete information and recommendations about recovery strategies comes from coaches and fellow athletes rather than peer-reviewed scientific publications (19). Therefore, to maximize recovery, and aid performance, it is important that coaches and athletes are educated and up to date on the types of recovery strategies that have been shown in multiple scientific publications to be effective, available, and best suited to athletes in training and competition.

With regards to recovery and training, the term *recovery run/ride/swim*, which is often used by aerobic endurance athletes to reflect an easy run, ride, or swim after competition, or a training session during the training cycle, must be clarified. Typically, a recovery session would fall under the banner of active recovery, which relates to low-intensity physical activities designed to facilitate recovery from the metabolic responses of exercise fatigue (72). A recovery session should be just that: a session with the intentional purpose of aiding recovery beyond that achievable via passive rest, which research indicates it does (40). It is not designed to provide an extra training stimulus and definitely should not contribute to any further muscle damage. Therefore, the recovery session should be short in duration and of a low intensity (i.e., zone 1 ([table 21.1](#))). Consequently, an extensive aerobic endurance session or long, slow distance session should not be used as a recovery run, just because it is of a low intensity. However, even though recovery sessions should not be considered as a training stimulus, they do contribute to the total amount of exercise completed in a training cycle and therefore must be included in all training plans for monitoring and recovery purposes (36).

TABLE 21.1 Example of a Five-Zone Intensity Scale to Prescribe and Monitor Training of Aerobic Endurance Athletes

Intensity zone	$\dot{V}O_2$ (% max)	Heart rate (% max)	Lactate (mmol/L)	Typical accumulated duration within zone
1	50-60	60-72	0.8-1.5	1-6 h
2	66-80	72-82	1.5-2.5	1-3 h
3	81-87	82-87	2.5-4.0	50-90 min
4	88-93	88-92	4.0-6.0	30-60 min
5	94-100	93-100	6.0-10.0	15-30 min

This scale is typical of intensity zone scales used for aerobic endurance training prescription and monitoring. The scale was developed by the Norwegian Olympic Federation as a general guideline based on years of testing of cross-country skiers, rowers, and biathletes.

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Step 3: Training Intensity

Central to causing training adaptations in the body is the interaction of training intensity and duration. Generally, the higher the exercise intensity, the shorter the exercise duration. Adaptations in the body are specific to the **intensity**, or effort expended during a training session. High-intensity aerobic exercise increases cardiovascular and respiratory function and allows for improved oxygen delivery to the working muscles (105). Increasing exercise intensity may also benefit skeletal muscle adaptations by affecting muscle fiber recruitment (85). As exercise intensity is increased, greater recruitment of type II muscle fibers occurs to meet the increased power needs. This training stimulus allows those fibers to become more aerobically trained, thereby potentially improving overall aerobic performance.

The regulation of exercise intensity is critical to the success of each training session and ultimately the entire program. An exercise

intensity that is too low does not stress the body's systems and induce the desired physiological adaptations, whereas an intensity that is too high results in excessive fatigue and a premature end to the training session. In either instance, the training session will be poor and ineffective.

The most accurate methods for regulating exercise intensity are to monitor oxygen consumption during exercise to determine its percentage of $\dot{V}O_2\text{max}$ and to periodically measure blood lactate concentration to determine the relationship to LT. If $\dot{V}O_2\text{max}$ monitoring is not available, exercise prescriptions can use heart rate, ratings of perceived exertion, metabolic equivalents (in clinical populations), or exercise velocity to monitor exercise intensity. Power-measuring devices are also frequently used by elite and top-level amateur competitors, especially in cycling.

Heart Rate

Heart rate is likely the most frequently used method for prescribing aerobic exercise intensity. The reason is the close relationship between heart rate and oxygen consumption, especially when the intensity is between 50% and 90% of **functional capacity** ($\dot{V}O_2\text{max}$), also called **heart rate reserve (HRR)**. HRR is the difference between an athlete's maximal heart rate and their resting heart rate (24). The most common means of regulating intensity using this method is to determine the specific heart rate associated with the desired percentage of $\dot{V}O_2\text{max}$ or the heart rate associated with the LT. For the greatest precision, this necessitates laboratory testing to identify these exercise intensities. If laboratory testing is unavailable, the individual's **age-predicted maximal heart rate (APMHR)** is often used as the basis for determining exercise intensity. While there are numerous APMHR equations, all of which are estimates and have been shown to have varying degrees of bias, the Fox equation ($\text{APMHR} = 220 - \text{age}$) appears less prone to under- and overestimation of HRmax in the general population (123). It is important to note, however, that the Fox equation is also susceptible

to individual variability with regards to its accuracy, having a documented standard deviation of 10 to 12 beats per minute (123). Refer to the sidebar [Target Heart Rate Calculations](#) for formulas and sample calculations for determining aerobic endurance exercise heart rate ranges using the **Karvonen method** and the **percentage of maximal heart rate (MHR) method**. The relationship between $\dot{V}O_2\text{max}$, HRR, and MHR is shown in [table 21.2](#).

TABLE 21.2 Relationship Between $\dot{V}O_2\text{max}$, Heart Rate Reserve (HRR), and Maximal Heart Rate (MHR)

% $\dot{V}O_2\text{max}$	% HRR	% MHR
50	50	66
55	55	70
60	60	74
65	65	77
70	70	81
75	75	85
80	80	88
85	85	92
90	90	96
95	95	98
100	100	100

Although the Karvonen and percentage of maximal heart rate formulas provide practical intensity assignments, basing them on APMHR will entail some inaccuracies (vs. laboratory-tested maximal heart rates) when exercise intensity is monitored (73). It has been determined that age contributes significantly to the variability in maximal heart rate; however, the effects of other factors such as mode of exercise and fitness level must also be considered with the use of heart rate to monitor intensity (73). Changes in fitness over

time must also be taken into account when prescribing exercise. As aerobic power ($\dot{V}O_2\text{max}$), LT, MLSS, and exercise economy improve, the prescribed absolute intensity will be a lower relative intensity, resulting in a lower stress and stimulus. As such, corresponding adjustments must be made to increase training intensity to maintain a stimulus that will elicit continuing and sustainable adaptations. To provide accurate information on the magnitude of these improvements in aerobic fitness, and to inform subsequent adjustments in training intensity, it is important to test athletes regularly in each of these factors. Refer to [chapters 13](#) and [14](#) for information on selecting, administering, and scoring appropriate validated field and laboratory tests. Additionally, using estimations of exercise intensity via estimated maximal heart rate equations provides no information about the intensity associated with the LT. Without some knowledge of an athlete's LT, achieved from field or laboratory testing, a highly effective aerobic endurance training program cannot be developed.

Target Heart Rate Calculations

Karvonen Method

Formula

- Age-predicted maximum heart rate (APMHR) = $220 - \text{age}$
- Heart rate reserve (HRR) = $\text{APMHR} - \text{resting heart rate (RHR)}$
- Target heart rate (THR) = $(\text{HRR} \times \text{exercise intensity}) + \text{RHR}$

To determine the target heart rate range (THRR), calculate THR using both the prescribed minimum and maximum intensity values.

Example

A 30-year-old athlete with an RHR of 60 beats per minute is assigned an exercise intensity of 60% to 70% of functional capacity:

- $\text{APMHR} = 220 - 30 = 190 \text{ beats/min}$
- $\text{RHR} = 60 \text{ beats/min}$
- $\text{HRR} = 190 - 60 = 130 \text{ beats/min}$

- Lowest number of the athlete's THRR = $(130 \times 0.60) + 60 = 78 + 60 = 138$ beats/min
- Highest number of the athlete's THRR = $(130 \times 0.70) + 60 = 91 + 60 = 151$ beats/min

Percentage of Maximal Heart Rate Method

Formula

- Age-predicted maximum heart rate (APMHR) = $220 - \text{age}$
- Target heart rate (THR) = $(\text{APMHR} \times \text{exercise intensity})$

To determine the target heart rate range (THRR), calculate THR using both the prescribed minimum and maximum intensity values.

Example

A 20-year-old athlete is assigned an exercise intensity of 70% to 85% of maximal heart rate:

- $\text{APMHR} = 220 - 20 = 200$ beats/min
- Lowest number of the athlete's THRR = $200 \times 0.70 = 140$ beats/min
- Highest number of the athlete's THRR = $200 \times 0.85 = 170$ beats/min

It is also important to note that environmental factors such as heat, humidity, and altitude place additional stresses on the body, affecting the cardiovascular responses to exercise (68, 117). As such, the acute and chronic effects of these external factors in elevating heart rate should be taken into account when prescribing and monitoring exercise intensity.

Ratings of Perceived Exertion Scales

Ratings of perceived exertion (RPE) scales also can be used to regulate and monitor intensity during aerobic endurance training. Evidence suggests that RPE can be used to accurately regulate intensity when there are changes in fitness level (29); however, researchers have demonstrated that the RPE–intensity relationship can be influenced by various external environmental factors such as passive distracters and environmental temperature (93, 130). According to Haddad, Padula, and Chamari (56), various subject characteristics such as age, sex, training status, and fitness level

may influence RPE. Furthermore, a few environmental factors that may influence RPE are listening to music, watching television or video, environmental temperature, altitude, nutritional considerations, and external feedback. However, these authors suggest that despite the potential influence of these factors, RPE is still a valid monitoring tool ([table 21.3](#)).

TABLE 21.3 Sample Rating of Perceived Exertion (RPE) Scale

Rating	Description
1	Nothing at all (lying down)
2	Extremely little
3	Very easy
4	Easy (could do this all day)
5	Moderate
6	Somewhat hard (starting to feel it)
7	Hard
8	Very hard (making an effort to keep up)
9	Very, very hard
10	Maximum effort (cannot go any further)

Metabolic Equivalents

Metabolic equivalents may also be used to prescribe exercise intensity. One **metabolic equivalent (MET)** is equal to $3.5 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ of oxygen consumption and is considered the amount of oxygen required by the body at rest (3). Metabolic equivalent values have been determined for a variety of physical activities; an activity with a MET value of 10.0 requires 10 times the oxygen uptake that is required by an individual at rest. Assigning MET values as part of an aerobic exercise prescription requires the strength and conditioning professional to know (or estimate) an athlete's maximal oxygen

uptake to be able to calculate an exercise MET level. However, the use of METs in exercise prescription is typically limited to clinical populations.

Intensity Measuring Technology

Previously due to cost, power devices for cycling such as power measuring cranks and hubs to monitor exercise intensity were mostly only used by professionals and top-level amateurs. Research has supported their use from the perspective of demonstrating that they provide valid and reliable power measurement (49). However, due to the advancements in technology, it is common to see athletes of all levels using technology to monitor exercise intensity. This technology ranges from smart watches to running stride power meters to power measuring cranks. Each technology has its own reasons for being selected. For example, using power measuring cranks in cycling has an advantage over other measures because metabolic rate is closely related to mechanical power production (30). Also, using power as an intensity measure allows reproducible intensity efforts regardless of environmental conditions, which may influence other measures of intensity such as heart rate and velocity (30). Irrespective of the technology selected to monitor power, speed, or similar variables, they should have training zones allocated to them that reflect the athlete's training status.

Technology also extends to advanced indoor trainers such as treadmills, rowers, and ski and cycle ergometers. Due to the large variety of technological devices currently available, it is not possible to address each manufacturer individually. However, some common points to consider are whether the measurements of power, speed, stride length, stride frequency, and the like are valid, accurate, and reliable. While a number of these commonly used devices have been research tested, many have not. Hence, it is important to review the product before buying. Also, given the number of variables that can be measured, it is important to set training intensity or target ranges before using them. Additionally, strength and conditioning

professionals must understand what each variable actually reflects, how to interpret it, and how it may change with respect to the training prescribed.

Similarly, while an athlete may be familiar with technology, it does not mean they are used to the specific device measuring their performance, and as such, a familiarization period may be required. For example, it can feel foreign to run on a treadmill (active, passive, or Alter-G) when one has not done so before. As such, coaching cues need to be developed and provided to the athlete to ensure he or she transitions successfully to the new technology or device. It also should be noted that variances in technology between indoor and outdoor equipment or differences in performance between exercising indoors versus outdoors exist and can produce differing results (90). For example, it has been shown that running inside on a treadmill at slower speeds can feel easier and produce a lower exercise heart rate than running outside at higher submaximal speeds (90). Aerobic endurance performance was also found to be superior outside compared to when performed indoors. The researchers suggest this may be related to a potential fear of falling from the treadmill at higher speeds and the potential build-up of heat due to the absence of air flow around the body when running indoors.

Training Zones

It is common practice in aerobic sports to prescribe exercise within one of a number of training zones, each of which reflects a training intensity and training goal, rather than a specific heart rate or RPE (62, 120). The number of zones can vary, typically from three to seven; however, the common reference point is often the athlete's lactate or ventilatory threshold. A low-intensity or aerobic training zone can be associated with a stable blood lactate concentration below 2 mmol/L with one or more high-intensity or anaerobic training zones above 4 mmol/L, and lactate threshold training can be seen as attaining and maintaining steady state lactate levels in the

zone between 2 mmol/L and 4 mmol/L (120). [Figure 21.1](#) and [table 21.1](#) ([page 697](#)) illustrate examples of three and five zone training models respectively.

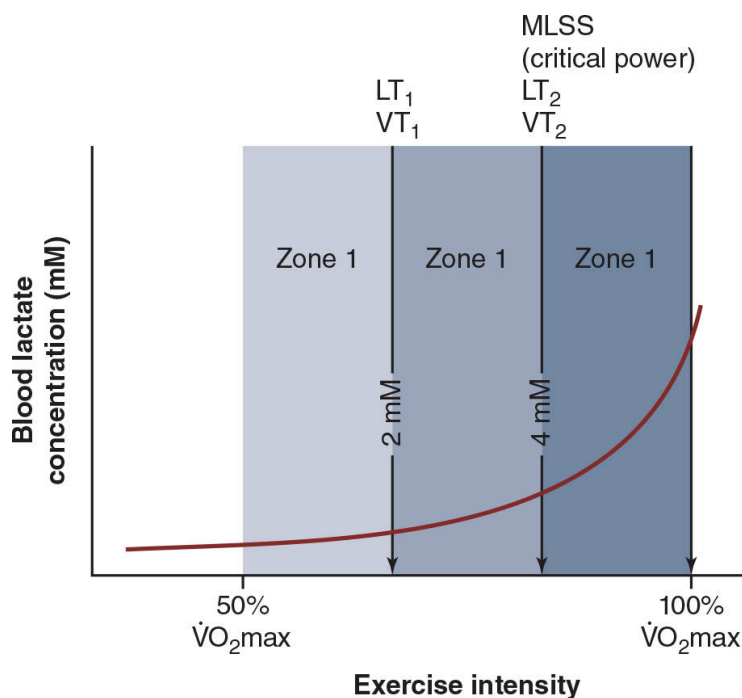


FIGURE 21.1 A three-intensity zone model based on identification of ventilatory thresholds. LT = lactate threshold; MLSS = maximal lactate steady state; mM = mmol/L.

Step 4: Exercise Duration

Exercise **duration** refers to the length of time of the training session. The duration of a training session is often influenced by the exercise intensity: the longer the exercise duration, the lower the exercise intensity. For example, exercise that is conducted at an intensity above the maximal lactate steady state (e.g., 85% of $\dot{V}O_{2\max}$) will have a relatively short duration (20-30 min) because the accumulation of lactate within the muscle will contribute to fatigue. Conversely, exercise that is performed at a much lower intensity (e.g., 70% of $\dot{V}O_{2\max}$) may be performed for several hours before the athlete experiences fatigue.

The duration of a training session is often influenced by the exercise intensity; the longer the exercise duration, the lower the exercise intensity.

Step 5: Exercise Progression

Once athletes begin an aerobic endurance exercise program, they need to continue training to either maintain or advance their aerobic fitness level. However, research indicates that aerobic fitness does not decrease for up to five weeks when frequency decreases to as few as two times per week as long as training intensity is maintained (126).

Increasing aerobic endurance fitness requires program progression, which initially involves increasing the frequency, intensity, and duration of exercise. Typically, for the general population, it is recommended that from week to week within a longer-term training plan, exercise frequency, intensity, or duration should not increase more than 10% each week to avoid overtraining or injury (57). While 10% may not seem like a significant increase, when these increases and their associated adaptations are added over a longer period of months and years, the adaptations and performance improvements are substantial and achieved while minimizing the risks of injury and overtraining. However, for athletes preparing for long-distance events, and especially those starting from a low fitness level, it may be necessary to adjust this guideline for a small number of weeks within the larger training plan. It should be noted that if an increase in training volume above 10% is implemented, it is critical that before doing so monitoring systems for physical and psychological overtraining symptoms are determined and implemented on a regular basis to recognize indications of overtraining and possible injury early on. For example, a cyclist riding 68 miles per week across three sessions may increase their 18-mile tempo and 18-mile interval sessions by 10%, thereby increasing each session by about 2 miles. Given the relatively high intensity of these training sessions, this would be appropriate.

However, to only increase the remaining extensive endurance session of 32 miles by 10% would only result in an increase of about 3 miles or, assuming they ride at an average of 18 miles per hour, an increase of 10 minutes, from 1 hour 46 minutes to 1 hour 56 minutes. Instead, given the low impact and intensity of exercise and the time involved, the duration of this extensive endurance ride may reasonably and safely be extended by 15 to 20 minutes (i.e., about 6 miles), which would equate to an increase of approximately 20%. But, as stated previously, this increase above the recommended 10% should be done cautiously, with injury and overtraining monitoring systems implemented, and only for a short period of time with an appropriate increase in recovery time and techniques reflective of the increased stress on the body. This type of planned increase in training frequency, intensity, or duration may be referred to as *short-term functional overreaching*.

An important point regarding training duration progression is that it is often recommended to use predicted time to set the duration of a training session rather than distance. This is because training time factors in the effects of terrain and environmental conditions, whereas distance does not. For example, if an athlete runs 10 miles over a flat course in 1 hour and 10 minutes, and then runs a hilly 11-mile course the following week in a much slower time of 1 hour and 58 minutes, even though the prescribed distance only increased by 10%, the time taken was considerably greater (+48 minutes = 65% increase) which is too much for a week-to-week progression.

Consequently, discretion needs to be used with respect to which training variable is increased and by how much. Further, to again avoid symptoms of overtraining or injury, only one training variable should be increased at a time.

Progression of training frequency may be limited by constraints such as school, work, and family commitments. It may not be possible for the athlete to train every day or to incorporate more than one training session each day. The duration of each training session is limited by the same constraints. In these cases, to elicit

further adaptations, a greater reliance on manipulating training intensity exists. Note that as the athlete's level of fitness changes over the course of the sport seasons, training intensity levels (e.g., heart rate range for a given training zone) must be adjusted. Similarly, as the training environment changes (e.g., from cool to hot and humid), RPE charts may be particularly helpful for monitoring intensity in addition to heart rate, which will be elevated by the increased physiologic demands induced by the environmental conditions. In conjunction with these points, progression of training frequency, intensity, and duration, along with recovery, should be monitored at all times to ensure appropriate periods of loading and unloading are applied, and signs or symptoms of overreaching or overtraining are detected early. More detailed information on these topics can be found in [chapter 24](#).

Types of Aerobic Endurance Training Programs

Several types of aerobic endurance training programs exist, each with varying frequency, intensity, duration, and progression parameters. Each type incorporates the five design variables and results in regimens created for specific outcomes. [Table 21.4](#) summarizes the types of aerobic endurance training and their common prescriptive guidelines. Sample training programs for each type of aerobic endurance training are included after the sections that follow.

TABLE 21.4 Types of Aerobic Endurance Training

Training type	Frequency per week*	Duration (work bout portion)	Intensity
Extensive endurance	1-2	Race distance or longer (~30-120 min)	$\leq 65\%$ of $\dot{V}O_2\text{max}$
Pace/tempo	1-2	~20-30 min	At the LT; at or slightly above race pace
Interval	1-2	3-5 min (with a work:rest ratio of 1:1)	Close to $\dot{V}O_2\text{max}$
High-intensity interval training	1	30-90 s (with a work:rest ratio of 1:5)	Greater than $\dot{V}O_2\text{max}$
Fartlek	1	~20-60 min	Varies between LSD and pace/tempo training intensities

*The other days of the week are composed of other training types and rest or recovery days or sessions.

Data from Lamb (1995); Penteado et al. (2014); Svedenhag (2008); Wells and Pate (1995).

Extensive Endurance Training

Many aerobic endurance coaches and athletes over the years have used the term **long, slow distance (LSD)** or high-volume training to reflect training for a long period of time at a slow pace. However, this term can give the wrong impression of what an athlete is actually trying to achieve—that is, to extend their aerobic endurance capabilities, not just exercise slowly for a long duration. Hence, the term **extensive endurance training** is a more appropriate term and the one used in this text. Irrespective of the term used, a common factor with extensive endurance training is that the training distance should be greater than race distance, or the duration should be at least as long as 30 minutes to 2 hours (98). However, as the race distance increases, it is not practical to complete an extensive endurance training session greater than the race distance. For example, because of the level of fatigue and tissue damage, it is not recommended for athletes to run 26 miles (42 km) or greater in training before running a marathon. What the athlete relies on in this situation is a combination of his or her extensive endurance runs plus the accumulation of the training volume over the weeks of training before the event. The length of an extensive endurance session will vary according to the training status of the athlete, where the athlete is in the program, the mode of exercise (e.g., cycling vs. running), as well as the duration of the aerobic endurance event for which he or she is training. Generally, as the distance increases from a short (e.g., 5,000 m) to a long (e.g., 26.2-mile marathon) or ultradistance aerobic endurance event, the concept of running the race distance or greater becomes less likely and unnecessary. However, it also should be noted that doing greater than the race distance at the lower end of the aerobic endurance distances (e.g., 5,000 m, 10,000 m, or short-distance triathlons) can be beneficial and is often undertaken.

An extensive endurance training session is typically completed at an intensity equivalent to training zone two (refer to [table 21.1](#) on [page 697](#)). This intensity is typically characterized as “conversation”

exercise, with the athlete able to talk without undue respiratory distress, or at a speed that the athlete feels they could run indefinitely. In general, increases in duration are prioritized when programming extensive endurance training sessions into a training program. It is important to not substantially increase the speed of the extensive endurance sessions in combination with increases in session duration, for as speed increases the stress of the session also increases, which in turn lengthens the posttraining recovery time and therefore can also increase the risk of overtraining or injury. The physiological benefits derived from extensive endurance training primarily include enhanced cardiovascular and thermoregulatory function, improved mitochondrial energy production and oxidative capacity of skeletal muscle, and increased use of fat as a fuel (15, 20, 64, 75, 83, 85, 96, 104, 136, 140, 142). These changes are likely to improve the LT intensity by enhancing the body's ability to clear lactate. Chronic use of this type of training also causes a change in the metabolic characteristics of the involved muscles (27) and an eventual shift of type IIx fibers to type I fibers (103, 124).

The increase in fat use also causes a sparing of the body's limited muscle glycogen reserves (60, 64, 69, 136). The intensity during extensive endurance training is lower than the intensity used during competition. However, extensive endurance training does not stimulate the neurological patterns of muscle fiber recruitment that are required during a race (136), and this may result in adaptations in muscle fibers that are not used during competition.

Aerobic endurance training should not be based solely on extensive endurance training but be combined with other training methods such as interval and pace/tempo training to train race pace endurance, and improve LT and exercise economy, thereby ensuring a complete aerobic endurance-trained athlete. Even elite distance running athletes recognize the benefit of this practice, running approximately 80% of total running distance at extensive endurance pace over the training year but also incorporating various other

types of speed training (59). This ensures the athlete is not only more aerobically efficient and has muscular endurance but also has race speed capabilities. It is also worth noting that extensive endurance sessions provide athletes with mental strength and stamina by experiencing extended periods of time exercising, which they will encounter on race day.

Pace/Tempo Training

Pace/tempo training employs an intensity at or slightly higher than race competition intensity. The intensity corresponds approximately to the maximal lactate steady state; therefore, this type of training is also often called **threshold training** or **aerobic-anaerobic interval training** (98). There are two ways to conduct pace/tempo training: steady and intermittent (98). Steady pace/tempo training is continuous training conducted at an intensity equal to the maximal lactate steady state for durations of approximately 20 to 30 minutes. The purpose of pace/tempo training is to stress the athlete at a specific intensity and improve energy production from both oxidative and nonoxidative metabolism. Intermittent pace/tempo training is also referred to as **tempo intervals**, **cruise intervals**, or threshold training (98). During intermittent pace/tempo training, the intensity is the same as for a steady-threshold workout, but the training session consists of a series of shorter intervals with brief recovery periods between work intervals. During pace/tempo training, it is important to avoid exercising at a higher intensity than the prescribed pace, assuming the prescribed pace is an accurate reflection of maximal lactate steady state. If the workout seems relatively easy, it is better to increase the distance than to increase the intensity. The primary objective for this type of training is to develop a sense of race pace and enhance the body systems' ability to sustain exercise at that pace. Pace/tempo training involves the same pattern of muscle fiber recruitment as is required in competition. The benefits derived from

this type of training include improved running economy and increased LT and MLSS.

Interval Training

Interval training involves exercise at intensities close to $\dot{V}O_{2\max}$. The work intervals should last between 3 and 5 minutes, although they can be as short as 30 seconds (3). The rest intervals for 3- to 5-minute work intervals should be equal to the work interval, thereby keeping the work-to-rest ratio (W:R) at 1:1. Interval training permits the athlete to train at intensities close to $\dot{V}O_{2\max}$ for a greater amount of time than would be possible in a single exercise session at a continuous high intensity. This type of training should not be performed until a firm base of aerobic endurance training has been attained (74). Interval training is very stressful on the athlete and should be used sparingly. The benefits derived from interval training include an increased $\dot{V}O_{2\max}$ and enhanced anaerobic metabolism.

High-Intensity Interval Training

High-intensity interval training (HIIT) is a form of training that uses repeated high-intensity exercise bouts above MLSS or critical $\dot{V}O_2$ interspersed with brief recovery periods (76). One high-intensity interval training protocol is **Tabata training**. This methodology requires athletes to train at an intensity high enough to result in exhaustion after 7 to 8 repetitions of 20 seconds of work with 10 seconds rest between exercise periods (129). HIIT has been shown to provide a number of benefits to aerobic endurance performance in recreational to well-trained athletes (53, 92, 137). Research has shown that HIIT may result in increases in $\dot{V}O_{2\max}$ in beginner (137) and trained athletes (92); however, in well-trained athletes the results are less clear (97). The observed improvements may be, at least in part, the result of increases in mitochondrial biogenesis and function contributing to increased aerobic metabolism (67, 79). HIIT has also been associated with improvements in LT (48, 92) and

exercise economy (48, 53), which have both been strongly associated with aerobic endurance performance. In cases where improvements in aerobic endurance performance or factors associated with this performance are similar from HIIT and traditional low-intensity aerobic training, HIIT provides an advantage by taking significantly less time to complete (23).

According to Buchheit and Laursen (22), for an optimal stimulus it is necessary for athletes to spend a total of several minutes within the HIIT session above 90% of the $\dot{V}O_2\text{max}$. Both short (<45 s) and long (2- to 4-min) HIIT intervals can be used to elicit different training responses. For example, shorter intervals with higher intensity may elicit positive changes in neuromuscular power and exercise economy, and as the work duration for a single exercise bout is increased, the energy contribution from fast glycolysis will likely increase along with blood lactate levels, which may stimulate improvements in lactate metabolism.

In performing HIIT, the appropriate amount of rest between repetitions is critical. If the relief intervals are too short, the athlete will be unable to put forth a quality effort on subsequent exercise bouts and also will be at a greater risk for injury. If the rest periods are too long, many of the benefits experienced from challenging the glycolytic energy system will likely diminish. An example of an appropriate work-to-rest ratio for long-interval HIIT training is ≥ 2 to 3 minutes at or above 90% $\dot{V}O_2\text{max}$, with ≤ 2 minutes of passive or low-intensity active recovery (21, 22, 77).

Fartlek Training

Fartlek training (the word *Fartlek* originates from the Swedish term for speed play) is a combination of several of the previously mentioned types of training. Although *Fartlek* training is generally associated with running, it can also be used for cycling and swimming. A sample *Fartlek* run involves easy running ($\sim 70\%$ $\dot{V}O_2\text{max}$) combined with either hill work or short, fast bursts of running ($\sim 85\text{-}90\%$ $\dot{V}O_2\text{max}$) for short time periods. Athletes can

apply this basic format to cycling and swimming by simply combining extensive endurance training, pace/tempo training, and interval training. A Fartlek training workout challenges all systems of the body and may help reduce the boredom and monotony associated with daily training. This type of training is likely to enhance $\dot{V}O_2\text{max}$, increase the LT, and improve exercise economy and fuel use.

The various types of training induce different physiological responses. Ideally, a sound program would incorporate all types of training into the athlete's weekly, monthly, or yearly training schedule.